

CASE STUDY

Finance Intelligence Platform

Automating financial reporting for Horizon Capital Partners — from raw spreadsheet to AI-reviewed, stakeholder-ready report, with a full audit trail at every step.

FastAPI

OpenAI

n8n Workflow

Airtable

65 Tests Passing

1 Client Problem

Horizon Capital Partners' analysts spent **2–3 hours a day** on manual, repetitive financial reporting work, with no reliable way to trace a headline number back to the transactions behind it:

- Cleaning and reconciling spreadsheets by hand
- Calculating KPIs manually, client by client
- Spotting anomalies by eye, easy to miss under time pressure
- Writing executive summaries from scratch each cycle
- Distributing reports to stakeholders manually

2 Objectives

Reduce the daily workload to under 15 minutes without sacrificing accuracy or auditability. The platform had to deliver, end-to-end:

- **Upload-triggered ingestion** — no manual pipeline kick-off
- **Validation** that catches data issues before they reach a report
- **KPI calculation** consistent with existing analyst methodology
- **Anomaly detection** matching what a careful analyst would catch
- **Decision-ready** executive summaries, not raw data dumps
- **Automatic notification** and a full audit trail for every figure

2–3 hrs

manual work replaced, per analyst per day

8

end-to-end success criteria defined upfront

1

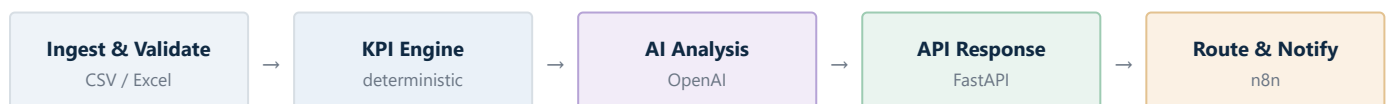
automated pipeline spanning 5 stages

100%

of figures traceable through the audit trail

3 Solution

A single automated pipeline that mirrors the analyst workflow it replaces, so behavior stays explainable and every stage is independently reviewable — a human can verify or override any step without re-deriving the whole pipeline.



Every uploaded file passes through all five stages before a report reaches a stakeholder — no step is skipped, and every step is logged.

4 Technologies Used



FastAPI

HTTP entry point & error handling



OpenAI

gpt-4o-mini · anomaly & summary generation



openpyxl

Real-world .xlsx parsing



n8n

Visual workflow orchestration



Airtable

Report archive & audit trail



Slack / Email

Severity-routed notifications



Python 3.11

Core language, typed throughout



pytest

65-test automated suite

5 Challenges Solved

Messy real-world spreadsheets

Merged cells, hidden sheets, and mixed date formats broke naive parsing. Solved with an `openpyxl`-based reader that resolves merges and skips non-visible sheets automatically.

LLM hallucination risk

Financial figures can't be left to a model's arithmetic. All KPIs are computed deterministically in Python first; the AI is instructed to narrate the given numbers, never invent or recompute them.

Flaky third-party AI calls

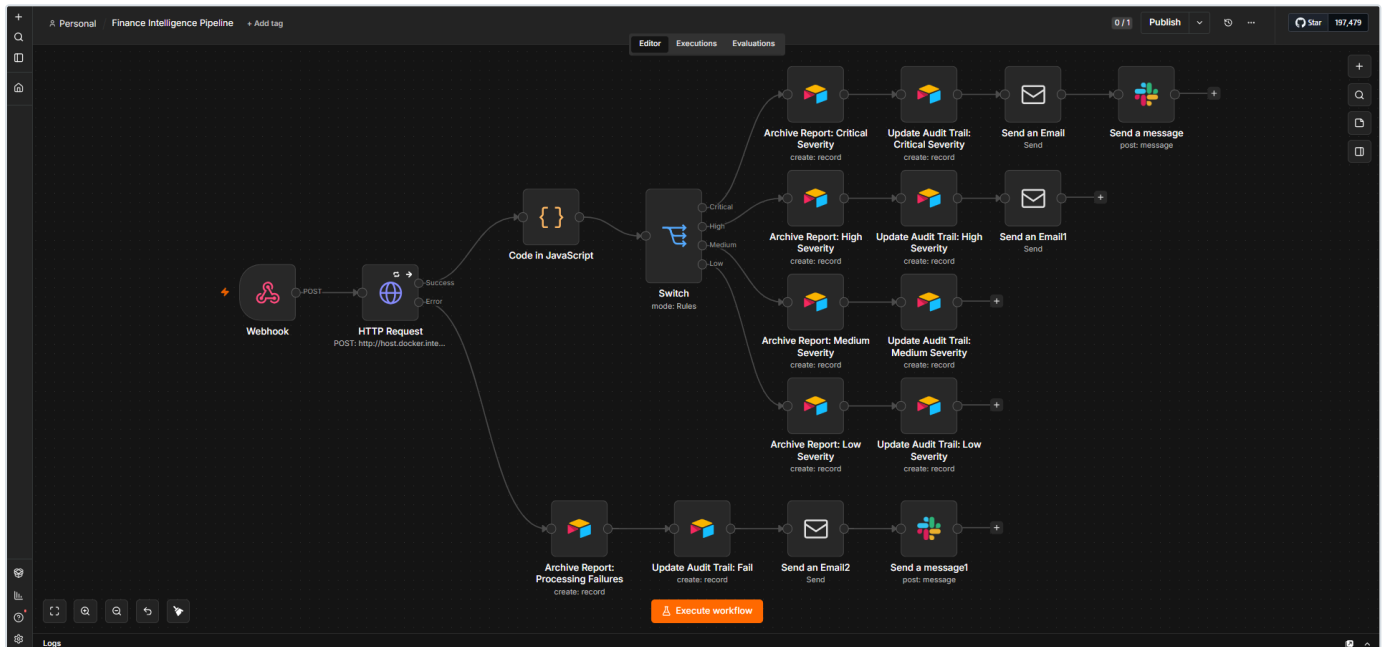
Timeouts and rate limits are inevitable at scale. Transient failures retry with exponential backoff; non-retryable errors (bad request, auth) fail immediately instead of wasting time.

Alert fatigue

Every report notifying every channel would drown out what matters. Severity-based n8n routing (low/medium/high/critical) sends each finding only to the audience and channel it warrants.

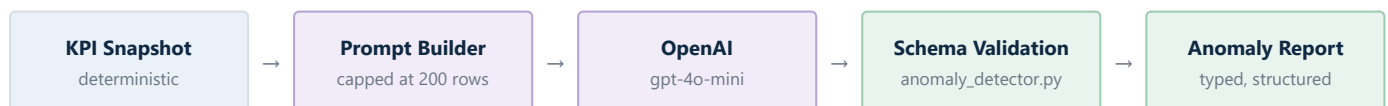
6 Architecture

One FastAPI service exposes the pipeline over HTTP (`POST /analyze`). n8n orchestrates everything downstream of a response: a `Switch` node routes each report by severity to `Airtable` (archive + audit trail) and `Slack/Email`, while a dedicated failure branch records processing errors instead of dropping them silently.



The imported n8n workflow: `Webhook` → `HTTP Request` → `Switch` → per-severity `Airtable` and `Slack/Email` nodes, plus a dedicated failure branch.

7 AI Workflow



Guardrails: the model is told never to invent or recompute figures; its response must match a strict JSON schema (severity, title, description, evidence, recommendation); malformed or incomplete responses are rejected outright — no silent partial reports ever reach a stakeholder.

8 Testing

The automated suite exercises every module end to end — ingestion, KPI calculation, prompt building, AI response validation, retry logic, and the API itself — with **no test making a real network call**; all LLM calls are faked or mocked.

Area	What's verified	Tests
Ingestion	CSV/XLSX parsing, validation, cleaning, deduplication, merged cells & hidden sheets	9
KPI Engine	Totals, per-client/category grouping, monthly totals, largest transactions	6

Area	What's verified	Tests
Prompt Builder	Section structure, schema documentation, truncation, currency formatting	12
Anomaly Detector	Response parsing/validation, malformed JSON, severity normalization	18
OpenAI Client	Retry/backoff, exhausted retries, non-retryable errors, exception hierarchy	11
API	Every endpoint and every error-to-status-code mapping	9

Complemented by 10 manual end-to-end scenarios verified through the FastAPI Swagger UI and the live n8n workflow (upload, validation failures, OpenAI timeout retries, the health endpoint).

9 Results

VERIFIED

65 / 65

automated tests passing, zero network calls

VERIFIED

4 tiers

severity levels auto-routed to the right channel

VERIFIED

0

raw tracebacks ever exposed to a caller

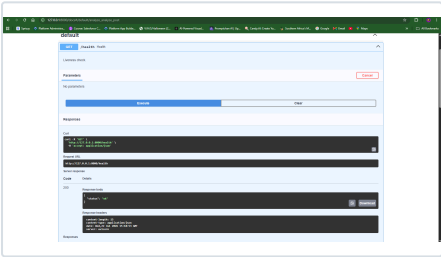
TARGET

<15 min

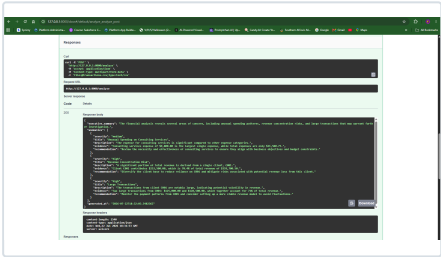
daily workload, down from 2–3 hours

The <15-minute workload target is the platform's business objective, not yet measured against live production volume — every other figure above is verified directly against the current codebase and test suite.

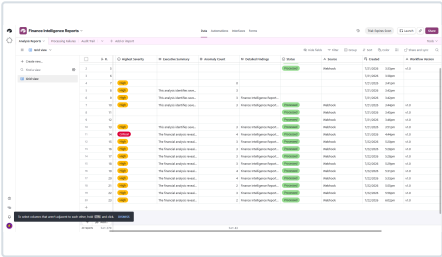
10 Screenshots



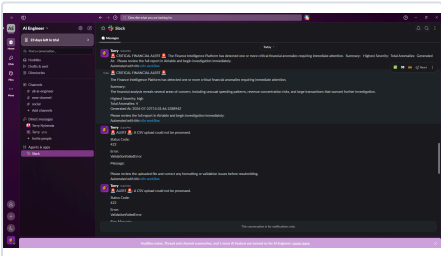
Health check — GET /health liveness probe



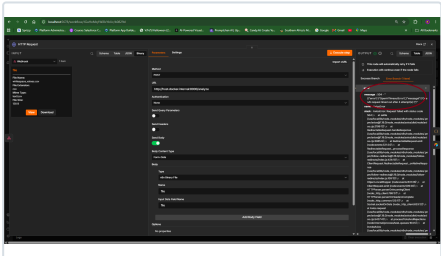
Anomaly report — POST /analyze response via Swagger



Report archive — Airtable analysis report table



Slack alert — severity-routed stakeholder notification



Retry handling — OpenAI transient failure recovering with backoff

11 Key Takeaways

- 1

Splitting deterministic math from AI narration keeps every figure in the report auditable back to its source transaction.
- 2

Guardrail prompting plus strict schema validation contain hallucination risk at a single, narrow boundary.
- 3

Severity-based orchestration turns one generated report into the right action for the right audience, automatically.
- 4

A fully mocked test suite proves pipeline logic correct without depending on a live, billed LLM connection.
- 5

Documenting what's built versus proposed keeps stakeholder expectations honest as the platform matures.
- 6

The architecture extends toward RBAC, batch processing, and horizontal scaling without requiring a rewrite.